# AI-Enhanced Reorder Point and Inventory Optimisation

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AI in Operations · Day 2, Session II exercise (slide 35): *"Enhance the reorder-point calculator and sensitivity analysis by introducing predictive intelligence, adaptive decision-making, and real-time optimization."*

## 1. The goal

A **reorder point** is the stock level at which a shop places a new order. The class calculator worked it out as *daily demand* × *lead time* + *safety stock*, with all three numbers typed in by hand. This project replaces the hand-typed numbers with values learned from real data, and replaces the one-at-a-time "sensitivity" charts with an AI search for the best decision.

## 2. The data

**FreshRetailNet-50K** (Dingdong, a Chinese online grocer; 2025; free licence CC BY 4.0): daily and hourly sales of fresh groceries with promotions, weather, holidays and hour-by-hour **out-of-stock records**. Following the slide's "no sharing of data with peers" rule, the notebook picks one store with a random seed made from my student ID: **store 308, 73 products**, 90 days of history (28 Mar – 25 Jun 2024) plus a final week (26 Jun – 2 Jul 2024) used as "new data". The forecasting models also learn from 25 other stores in the same city (2,499 store-product histories in total).

## 3. What was done

| Step | What it does | Key result |
|---|---|---|
| Find true demand | When a shelf is empty, recorded sales are lower than what customers wanted. Each product's normal hourly sales pattern is used to estimate the sales lost during empty hours. | Recorded sales understated true demand by **18.8%**. Tested on known days: **20.4%** error and almost no bias (–1%), vs **33.9%** too low if sales are treated as demand. |
| 1. Forecast demand | Six models compared on weeks they never saw: simple baselines, Holt-Winters (time series), Ridge regression, Gradient Boosting (machine learning) and an LSTM (deep learning). Inputs: past sales, weekday, promotions, holidays and weather. New raw data can be added, and forecasts update every day. | Gradient Boosting was best: **32.1%** average error on the final week vs **37.7%** for "same day last week". The LSTM forecast 27% too low, since 90 days is short for deep learning. |
| 2. Safety stock | The slide's formula, $Z \cdot \sqrt{(\sigma_D^2 \cdot L + D_{avg}^2 \cdot \sigma_L^2)}$, uses both demand and lead-time uncertainty ($D_{avg}$ = average daily demand). It is compared with a **Monte Carlo** simulation of 20,000 random futures, including supplier-risk scenarios. | Monte Carlo reorder points were on average **7.5%** higher than the formula's. The class calculator was on average **26%** lower. |
| 3. Bayesian optimisation | A simulator copies one product's shelf: random demand, random delivery times, and stock spoiling after 4 days. **Bayesian optimisation** then searches for the reorder point and order size with the lowest cost, under three limits: at least 95% of demand served, at most 5% waste, and limited storage. | It found the best policy in **40** simulations. A brute-force search needed **1,600**, and its best was 0.5% more expensive. Only 2.1% of the 1,600 policies tested meet all three limits. |
| Real time and checks | The final week is replayed day by day: new raw data arrives, the model re-trains, and the plan is re-optimised. A reorder plan is produced for all 73 products. A final section runs 44 automatic data checks. | The plan updated every day and stayed within all limits. All **44** data checks pass. |

## 4. Main result: focus product 201, tested on 300 new random futures

| Policy | Cost per day | Demand served | Waste | Within all limits? |
|---|---|---|---|---|
| Class calculator | 0.503 | 97.3% | 5.2% | No (too much waste) |
| Textbook formula + standard order size (EOQ, capped at shelf life) | 0.757 | 94.4% | 27.4% | No (far too much waste) |
| **AI: Bayesian-optimised** | **0.459** | **95.5%** | **3.9%** | **Yes: the only one** |

**Why the AI policy wins:** the reorder-point formula never sees spoilage or the cost of empty shelves. For fresh food, extra stock that cannot sell in time is thrown away. The optimiser balances empty shelves, storage, waste and ordering all at once. Its what-if tests also found two targets that **no policy can meet** (a 3-day shelf life, or serving 98% of demand with at most 5% waste), each confirmed by brute-force search. These are business decisions for managers, not problems a better formula can fix.

## 5. Assumptions and limits

- The data has no supplier information, so delivery time (1–4 days, average 2.15), costs and limits are stated assumptions. They can be changed in one place in the notebook.
- Sales are "scaled units": the company multiplied all figures by a hidden number, so results are relative, not real item counts.
- Only 90 days of history, so there are no yearly seasonal patterns. The future week's weather is not known in advance, so last week's weather is used.

**Files:** `Harsh_Verma_as25dxb024_AI_Reorder_Point.ipynb` (runs in Google Colab, estimated 10–15 minutes on the free CPU) and `replenishment_plan_store308.csv` (created by the notebook). Tools: Python, scikit-learn, PyTorch, statsmodels, Optuna.